

Recent Interview Questions

1. What is your role and responsibility in your current or previous project?

Answer:

I worked as a DevOps Engineer. My tasks included managing cloud infrastructure, setting up CI/CD pipelines, using Docker and Kubernetes, writing automation scripts with Terraform, and monitoring systems. I also supported application deployments.

2. What AWS services have you used in your project and how?

Answer:

I used EC2 for virtual machines, S3 for storing files, RDS for databases, VPC for networking, IAM for user access control, CloudWatch for monitoring, and Lambda for serverless tasks. I managed them mostly using Terraform.

3. How would you rate your Linux skills?

Answer:

I would rate myself 7 or 8 out of 10. I can work well with the terminal, run system commands, write shell scripts, and handle file permissions, services, and logs.

4. What are some common Linux commands you use regularly?

Answer:

I use commands like ls, cd, top, ps, grep, chmod, chown, df, and journalctl. These help me manage files, check performance, find logs, and handle user permissions.

5. How can you check CPU usage on a Linux system?

Answer:

I use top, htop, or mpstat to check CPU usage. top shows live usage of CPU, memory, and processes. htop is more user-friendly.

6. How would you find the last 5 users who logged into a Linux machine?

Answer:

I can use the last command to see recent logins. To see the last 5 users, I run:

```
last -n 5
```

7. How would you containerize an Nginx web application using Docker?**Answer:**

I would create a Dockerfile with nginx as the base image, copy the website files into the container, and expose port 80. Then I'd build and run the Docker container.

8. You have a Node.js app on port 8081 — how would you write a Dockerfile for it?**Answer:**

```
FROM node:18
WORKDIR /app
COPY . .
RUN npm install
EXPOSE 8081
CMD ["node", "index.js"]
```

9. What is a multi-stage Dockerfile and why is it useful?**Answer:**

A multi-stage Dockerfile helps reduce image size. The first stage is used to build the app, and the second stage copies only the final output. Yes, we can use the second stage as the base if needed.

10. Write a Dockerfile for a ReactJS application.**Answer:**

```
FROM node:18 as build
WORKDIR /app
COPY . .
RUN npm install
RUN npm run build

FROM nginx:alpine
COPY --from=build /app/build /usr/share/nginx/html
```

```
CMD ["nginx", "-g", "daemon off;"]
```

11. Have you worked with Kubernetes? Can you describe your experience?

Answer:

Yes, I have deployed applications using Kubernetes. I created Deployments, Services, ConfigMaps, and used kubectl to manage clusters. I also worked with Helm charts and monitored pods.

12. Based on a Dockerfile, how would you write a Kubernetes Deployment manifest?

Answer:

I would create a YAML file with apiVersion, kind: Deployment, and define the spec with replicas, container image, ports, and labels. Then apply it using kubectl apply -f.

13. What are the types of services in Kubernetes and when would you use each?

Answer:

There are 4 main types:

- ClusterIP (default, internal access)
- NodePort (exposes service on a static port on each node)
- LoadBalancer (uses cloud load balancer)
- ExternalName (maps service to an external DNS)

14. What is a headless service in Kubernetes and what's its use case?

Answer:

A headless service doesn't have a cluster IP. It's used when we want DNS resolution without a load balancer, such as in StatefulSets or when each pod should be accessed directly.

15. A pod is stuck in a CrashLoopBackOff state — how would you troubleshoot this?

Answer:

I would check logs using kubectl logs <pod-name>, then inspect the pod

using `kubectl describe pod`. It usually happens due to app errors or failed dependencies.

16. In a Kubernetes cluster, one ReplicaSet is not functioning properly. How would you debug it?

Answer:

I would describe the ReplicaSet and its pods using `kubectl describe rs <name>`. Then check if the pods are scheduled, healthy, and if there's any issue in the events section.

17. How does a Kubernetes application communicate with the external world?

Answer:

We expose the app using a Service of type LoadBalancer or NodePort. If it's behind an Ingress controller, we use an Ingress resource for routing external traffic.

18. What are the lifecycle states of an EC2 instance?

Answer:

The main states are:

- Pending
- Running
- Stopping
- Stopped
- Shutting-down
- Terminated

19. What is VPC Peering, and how does it work across regions or accounts?

Answer:

VPC Peering connects two VPCs so they can communicate privately. It works across regions and accounts using peering connections, route tables, and proper security group rules.

20. How do you connect to an EC2 instance inside a private subnet?

Answer:

I use a bastion host in a public subnet. First, I connect to the bastion host using SSH, then connect to the private EC2 instance from there.

21. What's the difference between Security Groups and NACLs in AWS?

Answer:

Security Groups work at the instance level and are stateful. NACLs work at the subnet level and are stateless. NACLs have separate rules for inbound and outbound traffic.

22. How would you set up VPC Peering between two VPCs?

Answer:

Create a VPC peering connection, accept it in the other VPC, update route tables in both VPCs, and allow traffic in security groups and NACLs.

23. What is the AWS Shared Responsibility Model? Can you explain it with an example?

Answer:

AWS handles security *of* the cloud (hardware, network), while we manage security *in* the cloud (data, access, configurations). For example, AWS secures the servers, but we must set up IAM roles properly.

24. How does S3 manage versioning and object locking for data durability?

Answer:

Versioning keeps all versions of an object. Object locking prevents deletion for a set time. These features help protect data from accidental loss or overwrite.

25. How would you monitor the performance of an EC2 instance using CloudWatch?

Answer:

I check metrics like CPU, disk, memory, and network usage. I also set alarms to notify when usage is too high or something goes wrong.

26. How can you use CloudWatch to track application metrics on EC2?

Answer:

Install and configure the CloudWatch agent. It collects logs and custom metrics like memory or app-level data. Then I view them in the CloudWatch dashboard.

27. Can you write a Terraform script to provision an EC2 instance and an S3 bucket?**Answer:**

Yes. I define resources like `aws_instance` and `aws_s3_bucket` in Terraform, then use `terraform apply` to create them. I also use variables for reusability.

28. Write Terraform code to create 5 EC2 instances, each with a unique name and instance type.**Answer:**

```
variable "instances" {  
  default = {  
    "web1" = "t2.micro"  
    "web2" = "t2.small"  
    "app1" = "t3.micro"  
    "app2" = "t3.small"  
    "db1"  = "t3.medium"  
  }  
}  
  
resource "aws_instance" "multi" {  
  for_each = var.instances  
  ami     = "ami-12345678"  
  instance_type = each.value  
  tags = {  
    Name = each.key  
  }  
}
```

29. What CI/CD steps do you follow during software development?**Answer:**

I start with code push to Git. A pipeline runs tests, builds the app, creates a Docker image, and pushes it to a registry. Then it deploys to a test or production environment.

30. How do you build and push a Node.js Docker image to Docker Hub using a CI/CD pipeline?

Answer:

In the pipeline, I add steps to build the Docker image using the Dockerfile, tag it, log in to Docker Hub, and push it using docker push.

31. How would you connect AWS Lambda with API Gateway for serverless deployment?

Answer:

I create a Lambda function and an API Gateway endpoint. Then I connect the API to the Lambda function so that HTTP requests trigger the function.

32. Do you have experience working with RESTful or GraphQL APIs? How have you used them?

Answer:

Yes. I've used REST APIs to integrate services, test them using Postman, and monitor responses. I also deployed backend services that exposed REST endpoints for frontend teams.

1. Introduction

2. What is AWS and what type of services you have used in AWS.

3. What is a role and user in AWS? Also, tell difference between them.

4. How can you create a s3 bucket from CLI and also how can you view the list of s3 buckets in AWS.

5. How you will add a role to a specific service in the AWS let's take EC2.

6. Your jenkins service is not running what troubleshooting steps would you follow.

7. Name some of the plugins you have used in your projects.

8. Explain about the ci/cd stages?

9. Name some of the default variables in jenkins.

10. What is the port number for nexus? How will you store the war files in the nexus repo.

11. What type of repository you are using, either group repository or a standard repository.

12. What is the command to print env variables in Linux, and also what are some default env variables.
13. What is file operator.
14. If a user is created what must be his default path?
15. What is the command to create a user and group in Linux.
16. What type of Linux distros you have worked with and what is the command to see the current Linux distro.
17. If you enter a command Hostnamectl what will be the output.
18. In Linux user is created there will be few default files created what are they, and also where the user details will be stored in Linux.
19. Linux command to see the disk free space.
20. Write shell script to print largest number out of 3 numbers.
21. What is the command to run the bash script.
22. What is the default Linux directory and file permissions.
23. How you will pass arguments in Linux. Write a bash script to pass the arguments.
24. What is #!/bin/bash what happens if we don't use this in the shell script.
25. What is diff between soft link and hard link and commands to create them
26. What is the command to run a python.py file in Linux.
27. What is docker?
28. What is the difference between CMD and ENTRYPOINT
29. Suppose you are running some process in container how can you see from Linux server without going into the container.
30. Command to create docker container using an existing image for example nginx.

2nd round Technical interview questions(f2f)

Brief Introduction

1. A user don't have the execute permissions for an script file. What are the ways you will check and assign.
2. A jenkins service is not running in the instance what steps would you follow?
3. How would you troubleshoot the Disk space issue in Linux explain in details with steps?
4. What type of file permissions are there what are they?
5. can't able to SSH into a server what troubleshooting steps would you follow?
6. Linux server is not coming up after reboot what steps would you follow?
7. Jenkins service is not able to access from chrome, what would be the reason?
8. Where the docker logs stored in the Linux?
9. What is a terraform state file and backend?
10. Write terraform code for an instance launch, also use variables.
11. How would you manage terraform state file in a team environment?
12. Terraform file is corrupted how would you troubleshoot?
13. Docker architecture explain?
14. Jenkins plugins, what is the purpose of pollSCM & Build periodically?
15. How would you build a jenkins pipeline manual and automatic explain?
16. Jenkins pipeline explain, and also write a basic pipeline syntax?

17. How we can store the credentials in jenkins securely and call them in the pipeline?
18. What type of RDS you have worked with?
19. Basic sql commands on MYSQL?
20. How would you push the images to ECR?
21. What is meant by prometheus & grafana?
22. In which linux file we will mention the prometheus configs to get alerts?
23. Where the logs will be stored for grafana and prometheus in linux?
24. what is composefile and swarm?
25. A crontab is failing how would you troubleshoot?

Client round interview questions

1. Introduction, project explanation, roles & responsibilities.
2. What is an EC2 and how do you launch it from UI and through code, explain?
3. If a server is down what are your troubleshooting steps?
4. How do you rate yourself on linux?
5. Describe the commands that are used for checking RAM, ROM, CPU usage, Memory. What is the command for checking the running processes on Linux.
6. Explain about the importance of filesystem? What type of filesystem you are using?
7. What is the path for storing the logs in linux? What type of logs will be store in the /var/log/messages?
8. Command to create user & Tell me some files which are in /etc directory.
9. Expalain about the entries /etc/passwd, /etc/shadow, /etc/groups files?
10. Where does the jenkins logs stored in the linux?
11. Suppose jenkins service is not running, what troubleshooting steps will you follow.
12. what is terraform? What is a terraform statefile, how do you manage terraform statefile it in the team environment.
13. Terraform statefile corrupted how do you troubleshoot?
14. Explain about terraform provider block & resource block, terraform lifecycle.
15. Which type of S3 storage areyou using currently & to reduce the cost which type of s3 will you use.
16. Suppose the terraform is running in the version 5.12 and whenever we apply changes in the code and run it, will the terrafrom runs on 5.12 or will it take the latest version.
17. write terraform simple code for s3 backend.
18. What is ansible? Explain the architecture of it and it's use case in the production.
19. Where we can store the secrets in the ansible?
20. Explain how you can secure the yml file and secrets files in the ansible?
21. Can we changes the ansible-vault password? If yes what was the command to change?
22. Tell me some of the ansible playbooks.yml file that you have used in the production.
23. Explain about the /hosts & /roles in ansible?
24. What is jenkins and why we are using it?
25. Tell me the jenkins stages that you have worked with?
26. what is the plugin that we used to run muliple branches in the git?
27. Name some plugins you have worked with
28. Jenkins is pipeline running slow and how do you troubleshoot it?

29. How can you change the default password in Jenkins?
30. Where do you store the passwords and usernames of SonarQube or Nexus in Jenkins securely?
31. How will you run the stages parallel in Jenkins?
32. Default port number of Jenkins?
33. What type of parameters are there in Jenkins?
34. How do you troubleshoot if the Jenkins service is down? And tell the commands to start, restart or enable Jenkins service?
35. What is Git and why are we using it?
36. What is Git rebase & Git merge?
37. What are your branching strategies in Git?
38. What is Git stash?
39. Tell me some commands you know in Git?
40. Do you have any idea on GitHub Actions, and yes what it means & how it works?
41. What type of services have you worked with in AWS?
42. To establish a connection between the service to service, how do you achieve it?
43. What type of RDS have you used in your current project?
44. What is Lambda and why are we using it?
45. What is DNS & Route53?
46. What are your achievements in your current organisation?
47. Have you learnt any tool or anything on your own recently?
48. Client explained about his project, roles and then work timings.

Festo India – Round 1 DevOps Interview Questions

What is CI/CD?

Write a Python script to create an EC2 instance and enable monitoring.

How do you integrate SonarQube into a Jenkins pipeline? (Explain step-by-step)

Suppose no team members are available and you need to create the whole infrastructure from scratch – from setup to deployment and monitoring – how would you achieve this?

Do you have any knowledge or experience in migration from one cloud or infrastructure to another? Explain.

How do you troubleshoot a Jenkins pipeline if it fails?

What branching strategy do you follow in your project?

Write a sample YAML file in Kubernetes (any resource of your choice).

Do you have knowledge or experience managing VMs and virtual machines? Explain.

Explain your project CI/CD pipeline in detail.

What is a Jenkins runner? Explain its role.

In our project/organization we use GitLab CI – since you have experience with Jenkins, how comfortable are you adapting to new CI/CD tools or technologies?

1.Explain Your Project Infrastructure

2. When user hit the your URL What will happen..!?

3. How Route 53 will connect..!?

4. difference between the Git fetch/ pull....!?

5. What is Git Stash....!?

6. What is Git Cherry-pick....!?

7. How to reduce the Docker file size..!?

8. What is the difference between the COPY / ADD....!?

9. What is the difference between the CMD / Entrypoint....!?

10. Did you initialize the SonarQube in your CI/CD pipelines....!?

11. How did you initialize/ configure....!?

12. What is the Backend....!?

13. If one instance is deleted unexpectedly how we will recover with Terraform....!?

14. what is provisions....!?

15. What is providers....!?

16. Can we configure the on-prem server in Terraform....!?

17. If we configure how will you..!?

18. What are your Roles and responsibilities..!?

19. What are the Kubernetes components....!?

20. Explain the Kubernetes structure..!?

Deployment strategy

What is the procedure for Deployments

How do you deploy the same app in sit uat prod

Secure the data

Did u worked on automation scripts for desktop apps or any

While configuring tools what is the issue u faced

Tell me about ur project

What is the method u follow for prod

Branching strategies , recommend a best Branching strategy

Gradle experience

What is the major issue u faced while deployments

What kind of applications u usually worked on like exactly what

Do u directly deploy to prod after sit uat completed

LTI

Deployment strategy

What is the procedure for Deployments

How do u deploy the same app in sit uat prod

Secure the data

Automation scripts for desktop apps

While configuring what is the issue u faced

Tell me about ur project

What is the method u follow for prod

Branching strategies

Gradle experience

What is the major issue u faced while deployments

What kind of applications u usually worked on like exactly what

Do u directly deploy to prod after sit uat completed

Wipro ==

Entire pipeline set up end to end pipeline

Aws vpc , iam Roles, ec2 ecs

Java end to end Integrate end to end pipeline using aws terraform

Harness deployments code

Imperative vs declarative

How to integrate tools pipeline in jenkins

Explanation pom.xml along with what are the files exists in pom

Docker containerization

Git vs github

Github actions

Git merge Git rebase

Docker File Components

Docker namespaces

Mphasis ...

Self introduction

Declarative pipeline vs scripted pipeline

Mvn clean install mvn deploy mvn package

Parallel pipeline

Blue green deployment

Health checks

Master slave architecture

Shell scripting *

Lloyds

List running process

How to edit file and save in linux

Blue green deployment

Softlink and hardlink

Impetus Technologies:

Round-1:

1. How to automate a process using pipeline?
2. Types of errors you will face in pipeline?
3. How to create resources in terraform?
4. How to delete unused containers in docker?
5. Write a command to check utilisation of a server?
6. Write a script to find disk utilisation of a container?
7. How to integrate pod monitoring tools in Jenkins pipeline?
8. How to get notifications about pipeline?
9. Tell me the difference between deployment and stateful set?
10. What is use case for deployment and stateful set?
11. What type of deployment strategy you used in your project?
12. How to deploy an application in production?

Round-2:

1. How do you deploy an application in production environment? Explain.
2. What type of issues you faced while deploying an application in production environment? How did you overcome those issues?
3. When ever you updated the application with new deployment. how did you handle/route the traffic to new version application?
4. How you maintain code in Github?
5. Explain deployment strategy you used in your project?
6. How do you automation deployment?
7. How do you implement argoCD in deployment?
8. What is major issue in production you have faced in last 3 to 6 months?

EY:

Round-1:

1. How do you automate process using Jenkins pipeline?
2. How do you use (integrate) tools in Jenkins pipeline?
3. How do pass variables to pipeline?
4. How do you pass credentials in pipeline?
5. What is SCM?
6. What is use of branches?

7. What is merge and rebase? What is difference between merge and rebase?
8. How do you bring git hub code to local?
9. What is git push?
10. What is git stash?
11. Who do you trigger pipeline when code is committed in git?
12. How to create a file in Linux?
13. Write a command to create a directory?
14. How to change file permissions?
15. How to create user?
16. How to change owner to a file?
17. Give me the value for read permission and write permission?
18. What is ansible?
19. What is use of playbooks?
20. Write a playbook to install 3 services.
21. What is Kubernetes?
22. Explain Kubernetes architecture.
23. What is purpose of scheduler?
24. What is deployment? Explain a use case.
25. What is stateful set? Explain use case.
26. What is difference between deployment and stateful set?
27. What is docker file?
28. Explain docker components.
29. Explain docker file components.
30. What is difference between COPY and ADD?
31. What is nginx port number?
32. Write a docker file to a web-server and the image should be light weight.
33. What is docker volume and how do you attach a volume to a container?
34. Explain terraform. Tell me the command in terraform
35. What is state file?

1) Self-Introduction

- 2) In your project, What Linux commands will you use when you start your day ?
- 3) How do you check live system logs or errors?
- 4) How do you check network connectivity between servers?
- 5) How to integrate jenkins with ansible?
- 6) If jenkins build fails, how will you troubleshoot or resolve?
- 7) What jenkins stages you implemented in your project?
- 8) What are DDL & DML commands in SQL?
- 9) How will you find duplicate records in a table?
- 10) What are the roles and gathering facts in ansible?
- 11) What is idempotency

in ansible?

12) What deployment strategy you implemented in your projects?

13) What types of automation scripts you wrote and explain them

14) What challenges you faced while deployment and how you resolved?

15) Are you familiar with dynatrace and autosys?

16) What monitoring tools you worked and how did you integrate them with kubernetes?

How do you establish and check the connection between two servers.

Your server is not starting properly, how do you debug that.

What are the playbooks you have written.

Ansible vault

What type of Jenkins pipeline you built

What are the stages in it.

What is meant by a job in Jenkins

Your jenkins job is failing, how do you debug.

Why kubernetes?

How do you debug a Pod which is going to error state

How do you check/fetch error logs in a server

In which location a server normally stores the logs

Which type of deployment strategy are you following and why

Write a shell script that you have written in your project

Git branching strategies

How do you integrate Prometheus with Grafana

Why splunk is needed.

Devops Interview questions

1. Explain Devops tools

2. Differences between Docker and Kubernetes

3. What are docker file components and write Docker file?

4. What is Docker swarm?

5. What is Docker Compose?

6. What is kubernetes?

7. Differences between statefulset and Deployment?

8. What you mean by pod notready state?

9. What is ingress

10. Tell me about configmaps and Secrets?

11. Command for disk space

12. Command to check memory space

- 13.If your pod is starting continuously what you will do?
- 14.Command to go inside the pod?
- 15.What is 777 permission means?
- 16.What is the command to give write permission
- 17.what will be the command to check the largest file in the directory?
- 18.What is TLS?

How do you build a docker image?

What are different services that you work in AWS?

What is Route53? And how you do mapping in Route53?

What is application load balancer?

Which record we use for ALB in Route53?

What are the common problems that you troubleshoot in jenkins?

What troubleshooting you have done in Terraform?

Troubleshooting in k8s?

Have you created deployment file in k8s?

Interviewer asked me to open deployment yaml file in Google and asked to explain it.

If the pod is not coming up how to troubleshoot it?

How many types of services in k8s?

Difference between services in k8s?

What is difference between ALB and NLB?

How can we expose the service?

How can we expose using ingress?

He asked me to explain basic ingress yaml file

What are config map and secrets?

Deployment of any service in k8s

What is private and public subnet?

What is NAT gateway?

What is Internet gateway?

Commands to check cpu, memory and disk space

To update the jenkins what things you take care, take backup

Jenkins deployed in ec2 where the config info is stored

EBS

Interview -1

Explain Terraform provisioners

What is azure directory

What is azure entra ID

Write a nginx conf file,Suppose if you have 2 urls ,one is http request and another is https request,

Your traffic should be route to http

What is Ec2 instance and volume

Sopra Steria

JENKINS

1. How to integrate sonarqube with jenkins , how store values, implementation of sonarqube?
2. How to store values by using jenkins?
3. How many fields are using jenkins?
4. How to give permissions to users? have you ever add user?
5. Explain about Maven life cycle?
6. Explain about POM.XML file?
7. How to secure the jenkins project?
8. Explain About Pipeline Stages?
9. How to take backup by using jenkins?
10. In which components used by using jenkins?
11. In which plugins are used by using jenkins?
12. What is CI&CD ? Diff b/w Continuous Delivery and Deployment ? particularly in which time we are used? For the Which tool used for CD?
13. Explain pipeline stages? How many types? How many are you used? What is Diff B/w Them?

GIT

1. Diff b/w merge and rebase?
2. What is Sthash? Tell me about merge conflicts?
3. Tell some commands in git?
4. How to took the data from the code GIT PULL& GIT PUSH
5. How to integrate Webhooks? And how to take the code in by using JEnkins?

Docker

1. Explain about docker file components?
2. Tell me about Docker-compose File?
3. Docker container create command ? and How to go to inside container Command?
4. Explain the Docker ? Why its used?
5. For taking the images which you are using(Docker Hub) ? And command?
6. Why you are using Docker Hub? And some other tools are Available ? Particularly why?

K8s

1. What is K8S and tell me about archestecture?
2. Diff B?/w Secrets AND config maps?
3. Dif B/w Depolyment&Satefulset&Demonset?
4. Diff b/w POD & Deployent?
5. Why we are used Blue-green Deployments ? Explain?
6. Explan About Auto Scaling?
7. Waht is POD?

TERRAFORM

1. What is Terraform? Explain About IAAC?
2. By using terraform Which resource are creating ?
3. Diff B/w Security group & NACL?
4. What is SEcurity Groups? Explain About Ingress& Egress?
5. Tell me about Creation of Vpc?
6. Explain the Providers? Which Provider Are you using ? Tell me some Providers Names?
7. Tell me about Output Section?
BASH
8. Have you used Bash And Tell me how to write (what you are writing in this)?

Ansible

1. What is Ansible?
2. What is Play-books? Explain Play books?
3. What is Ansible Modules? Explain?
4. In betwween asking webserver something? How to start? And Stages?
Linux
5. Tell me some commands In whatever you know in Linux?
6. Which Operating System USed ?
7. How to write or secure the files By using Linux?
8. Asked Grep Commands?
9. you have Fie inside the folder how to give 755 permission?
10. How to see the file and inside the files to take That?

#pwc company:

◆ Bash Scripting

Write a bash script to find the top 5 largest files in a directory and display their size.

◆ Python Scripting

Write a Python script to read a log file and count how many times the word "ERROR" appears.

◆ AWS

Explain the difference between EC2, ECS, and EKS. When would you use each?

Difference between Security Groups and NACLs.

How does IAM role differ from IAM user?

What happens if an EC2 instance is stopped vs terminated?

Difference between ALB, NLB, and CLB.

How does Auto Scaling work with ALB?

What is VPC peering and its limitations?

Difference between S3 and EBS.

◆ Git

What is the difference between git merge and git rebase?

How do you resolve merge conflicts?

◆ GitHub Actions

How does a GitHub Actions workflow get triggered?

Explain the structure of a GitHub Actions YAML file.

◆ GitLab CI/CD

Difference between GitHub Actions and GitLab CI/CD?

What are stages and jobs in GitLab pipelines?

Explain the structure of a .gitlab-ci.yml file.

What are stages, jobs, and pipelines in GitLab?

Difference between artifacts and cache in GitLab.

How do GitLab runners work?

What is a protected branch and protected variable?

How do you trigger pipelines manually in GitLab?

◆ Docker

Difference between Docker image and container.

What is a multi-stage Docker build and why is it used?

◆ Kubernetes

Difference between Deployment, StatefulSet, and DaemonSet.

How does Kubernetes perform rolling updates?

How does Kubernetes handle pod restarts and failures?

What happens when a node goes into NotReady state?

Difference between ClusterIP, NodePort, and LoadBalancer services.

How do liveness and readiness probes work?

What is the purpose of ConfigMaps and Secrets?

How does Horizontal Pod Autoscaler work internally?

Difference between Ingress and Service.

What is the use of namespaces in Kubernetes?

How do you troubleshoot a pod stuck in CrashLoopBackOff?

◆ Terraform

What is the Terraform state file?

Difference between terraform plan and terraform apply.

#TechM

AWS

What is Auto Scaling and how do you configure scaling policies?

How to optimize cost for EC2, RDS and S3?

What is VPC and difference between Public/Private subnet?

How do you secure S3 buckets?

How does Load Balancer work in multi AZ setup?

Jenkins

Explain pipeline as code?

What stages you use in CI/CD?

How do you trigger pipeline automatically?

How do you store and use credentials in Jenkins?

What is Blue-Green deployment?

Docker & Kubernetes

Difference between Docker image and container?

What is multi-stage build in Docker?

Explain pod, deployment, service?

How to troubleshoot pod CrashLoopBackOff?

What is Ingress and how it works?

Terraform

What is IaC and why Terraform?

Explain Terraform state file?

How do you manage multiple environments (dev/qa/prod)?

What is the use of variables and outputs?

How to handle Terraform remote backend?

1. How do you decide to roll back or apply a hot-fix when a production issue occurs?

2. How would you respond to a production outage during peak hours?

3. How do you stop broken code from reaching production?

4. So suppose you are working on a project, I being your new client asks for a scalable, highly reliable, zero down time architecture model for my project, how do you design the architecture and what are the best practices and tools you use ?

5. Can we write a Dockerfile without a base image?

6. How can you shorten a CI/CD pipeline that currently takes 45 minutes?

7. Write a Bash script or command to get the total number of lines in a file.

8. How do you use multi-branch pipeline in you project ?

9. What types of issues have you handled through JIRA? Can you explain one major issue and one minor issue in detail, including your approach and resolution?

10. How well do you handle high-pressure situations? Can you share an example of a stressful incident you managed and how you approached it?

11. How do you design alerting so that you avoid noisy or false alerts?

12. How do you trigger a pipeline if:

a. Code is pushed to a specific branch?

b. Ignore if it is pushed to some other branch?

c. Trigger if a PR is raised?

*****AWS Networking & VPC*****

Difference between Public EC2 and Private EC2

What is a NAT Gateway, where does it reside, and how is it used?

Difference between NAT Gateway and Bastion Host

How does routing work for NAT Gateway?

What is an Internet Gateway?

How to access S3 from a private subnet without using public internet

Types of Load Balancers in AWS

What is a VPC

How to connect multiple VPCs together

Alternatives to VPC Peering

What is CIDR in AWS

How to calculate IP addresses from CIDR

How many IPs are reserved/blocked by AWS

Which IP addresses are blocked by AWS

*****Networking Fundamentals*****

Difference between TCP and UDP

Difference between HTTP and HTTPS

What does 'S' stand for in HTTPS

What is TLS/SSL

Who issues SSL certificates

How does the SSL/TLS Handshake work (high-level)

*****Docker & Containers*****

Explain Docker Architecture

Difference between Containers and Virtual Machines

How containers use OS resources

How resource allocation works for containers

Explain Dockerfile

*****Kubernetes*****

What are Services in Kubernetes

Types of Kubernetes Services

What is Ingress and Ingress Controller

How do you set up an Ingress Controller

How does traffic flow from domain → Ingress → Service → Pod

Difference between LoadBalancer Service and Ingress

*****Monitoring & Observability*****

What monitoring tools have you used?

What is Grafana and how do you configure it?

What is Prometheus

What are the components of Prometheus

How did you configure Prometheus.

Tell me your introduction & day-to-day activities?

2. Which cloud are you using & which AWS services are you using?

3. Tell me the difference between Public Subnet vs Private Subnet?

4. What is a namespace in Kubernetes?

5. What is the role of the state file in Terraform?

6. What are the most useful Terraform commands?

7. Have you worked on Terraform? How do you manage the state file? What is the Terraform architecture you are using?

8. Same Terraform file: if two people are working on it, how do you know if someone has created resources or not?

9. Suppose you are working on a Terraform script for EC2 infrastructure creation. A team member makes changes in the AWS Console, but your Terraform template is different. How do you fetch those changes into your Terraform template?

10. What is your application technology stack?

11. Which CI/CD tool are you using?

12. Tell me step by step, from basic to end level, how you deploy your Java application on an EC2 server?

13. How do you connect to an EC2 instance?

14. In Terraform, how do you connect to ECR?

15. In Kubernetes, what is RBAC?

16. How do you monitor the logs for pods running in an EKS cluster?

17. Are you using volume mounts? Using volume mount services or without volume mounts?

18. What is PVC?

19. What is ADD in a Dockerfile? What is the difference between CMD and ENTRYPOINT?

20. Tell the command in Ubuntu: if a process is running, how do you kill that process?

DP F2F Interview – DevOps / SRE | Interview Questions

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◆ Level 1

▪ How do you configure Jenkins?

▪ How does Blue/Green deployment work?

▪ Is there any downtime in Blue/Green deployment?

▪ A user is unable to access an S3 bucket – how do you troubleshoot?

▪ How do you design a highly available Kubernetes cluster?

▪ How do you serve 50% traffic to Team A and 50% to Team B in Kubernetes?

▪ What are your roles and responsibilities as a DevOps Engineer?

- Where do you deploy applications—on-premises or in the cloud?
- Write Terraform code to create an S3 bucket
- What is a recent Kubernetes issue you troubleshot?
- Which command is used to check pod logs?
- How does ArgoCD work?
- Explain the end-to-end CI/CD workflow using Jenkins
- How do you troubleshoot a server that is not working?
- What automation have you implemented using Ansible?

◆ Level 2

- How do you list Kubernetes pods based on age?
- What is the Terraform workflow?
- If a Terraform project has 100 resources, how do you create only one?
- Differences between DaemonSet, StatefulSet, and Deployment
- What is a headless service and how does it work?
- What are Kubernetes probes, and how many types are there?
- Can we create a StatefulSet without PV and PVC?
- Can we attach PV and PVC to a Deployment?
- How do you provide S3 access to an EC2 instance?
- What is Backstage and how does it work?
- What is APM (Application Performance Monitoring)?
- Do you worked on Observability, Monitoring and Tracing?

Please find the below TCS interview experience ,

Technical Round-1

1.how to integrate S3 with Jenkins

2 write Jenkins pipeline for your project and explain

3 write a Multistage Dockerfile for Java application

4 tell me the docker commands to create an image and Container

5 tell me about deployment strategies in K8S

6 explain Blue-Green deployment

7 explain about Ingress how its work internally in depth

8 monitoring dash board and logs looks healthy but when user try to access the application getting 500 bad gateway. what could be the reason

9 write a YML file for Service

10 explain your project

11 what is the use of state file and taint

Technical Round-2

1.how to integrate S3 with Jenkins

2 write Jenkins pipeline for your project and explain

3 explain about Docker compose and use

4 what the relation b/w deployment and pod and RS

5 difference b/w readiness and liveness probs

6 explain how the rolling updates works

7 how will u reduce docker image size

8 write a SQL query to fetch the EMP details from a table

Company: *Copart India*

Working on AWS Route53 and how do I host my own site on AWS (Currently it's hosted on godaddy)?

In a linux system I am a root user but still I am unable to create a file or a folder. Why?

Company: *PWC*

1st round:

What is RBAC and why do we use it and how?

What is the realtime difference between config maps & secrets

How many micro services did you work on?

Where do we store the credentials in jenkins and how

Explain stages in your pipeline

What is ASG in AWS

2nd round:

What is user data in AWS

I created a EC2 from the console (UI) now once that is active and ready then it should automatically perform some commands. Let's say it should install a tool the moment it's ready. How do we do that?

How do i store a file from my EC2 to S3

Storage classes in S3?

What is VPC peering in AWS?

What is the terraform version you are using in your project?

What are data sources in terraform?

Do you know how to perform the cluster upgrades in EKS?

S3 Life Cycle?

What are terraform workspaces?

How are two services communicated in AWS?

If i change anything on AWS console then will terraform know it if yes then how?

I am trying to launch an EC2 but I am unable to do it. How do you deal with it?

How do I know if my POD is ready or active?

What are some common Kubernetes errors?

If a POD is failing how do you troubleshoot it?

How do 2 applications communicate with each other? Maybe containers or 2 systems or pods but how are they communicated?

How are pods/containers communicated in kubernetes ENV

Lets say i want to install a tool in 100 EC2's whats your approach?

Which CICD tool did you work on and why?

Did you work on any deployments? If yes then explain what they are? And how?

What is cloudFront?

Explain Jenkins stages in your previous project briefly ?

How do you store any sensitive information, let's say passwords or any other imp info about your Cloud Infra?

Do you know about CDN? Or cloudFront in AWS

Company: EY

What is the difference between git merge & rebase

What is a zombie process

How is a shell script executed

What is -e in shell scripting

How do we integrate shell scripting in python

Write a python program you know

What are readiness & liveness probs

In linux if my server is using much resources then how do you do it

What are deployment strategies

What is a rolling update

What is terraform state file lock

Difference between for each & count in terraform

What are docker volumes

What is jenkins shared library

Why do we need nodes in jenkins

If a pod is having much latency how do you troubleshoot it

If your Jenkins pipeline is taking much time to run then how do you make sure it runs in a minimal time? Run on a different node?

Difference between declarative pipeline and scripted pipe line

What is inheritance in python

What are mutable and immutable objects in python

Company: Grid Dynamics

Where is the cached data stored in a browser

What are python environments | environmental variables in python

How do i see what ports are listening in linux

What are lists & tuples in python

How do you design a CI/CD pipeline for microservices vs monoliths applications?

How do you structure terraform code for multiple environments?

Pod is running and healthy but service is unreachable. Why?

Difference between NACL & SG Why and where do we use it?

Tell me about your daily usage of terraform commands

Terraform state file locking

Terraform modules

How do you handle secrets in terraform?

Diff between ansible & terraform why use it and where tell me with example?

Whats the use of terraform import?

Can you design me a CI/CD pipeline for 1 micro service?

What is terraform backend?

AWS DR concepts what do you know about AWS DR?

How much exposure do you have over DB's and what are they?

Diff between relational DB & non relational DB

What happens if one AZ fails ?

If disk is full how do you troubleshoot it?

Disk is free but i cant perform any actions why?

Whats the use of "free"

What are IAM roles users & policies ?

If i have to restrict access to someone over cloud how do you do it?

What are the types of instances that you know?

How AWS Billing works?

What are the AWS payment plans?

How do you design multi region architecture in cloud

Why should we design multi region architecture

Diff between HPA & VPA

What is high availability quote me with an example

How do we achieve fault tolerance and high availability in AWS?

What are the types of LB's you know and tell me when to use what?

How do you design a VPC?

Give me an examples in realtime where and how a VPC is used?

Explain me about few of the core components in VPC

How are pvt subnets & public subnets designed

How do you handle loadspikes in AWS? Quote me with an example

How do i achieve the communication between one VPC & other explain me the architecture

Can we replicate data in S3? If yes then give me an example and how you design it

When i launch an AWS Instance it should perform all the pre defined actions how do i achieve it?

What is the use of launch template and explain me how its designed

What is ASG and how its designed and quote me an real time example of how when and where to use it?

Diff between S3 EFS & EBS when and how to use it.

What security tool do you know in AWS to protect our network ?

Tell me the commands you know in docker?

How do you reduce a image size and why do we do that

Diff between pod replicaset & deployment

What happens when a container exceeded its given memory limit

When do you use multiple clusters vs namespaces?

What happens when a node goes down?

What are the debugging commands you use in kubernetes and why?

Tell me which deployment strategy you worked on and tell me the best one to use and why?

Any roll back strategies you are aware of ?

What are the top linux commands you used?

When i face networking issue what commands you would think of in linux?

When i face memory constraints in linux what commands you would think of?

What are the concepts you are aware of in python?

What kind of shell scripting commands you worked on?

All the best----- Shiva Sai.